StudentPay Escrow

Trustless Peer-to-Peer Student Micro-Services Escrow on BotChain Mainnet

Live DApp: https://student-pay-drik.vercel.app | Contract: 0x59540a9B86c2bFB4F2591c840A6377484200d680 | Chain ID: 677

1. The Problem

Students frequently exchange services with peers for tutoring, design work, software development, video editing, essay proofreading, and campus events. Traditional agreements carry high risks: buyers paying upfront risk sellers disappearing; sellers delivering work risk buyers withholding funds; and digital assets are often stolen prior to payment. Traditional escrow solutions charge 10-20% fees and rely on centralized servers.

2. The Opportunity

Higher education campuses host thriving micro-economies. A non-custodial, zero-commission escrow application enables students to monetize skills safely without trusting unknown counterparties or relying on centralized banking intermediaries.

3. The StudentPay Solution

StudentPay eliminates counterparty risk using EVM smart contracts on BotChain Mainnet:
• Zero Centralized Backend: The smart contract handles 100% of business logic and state.
• Blinded Proof Commitments: Sellers submit keccak256 hashes before deadlines to prevent work-stealing.
• Protected Preview Inspection: Buyers inspect watermarked previews (STUDENTPAY • PREVIEW ONLY) while usable originals remain locked.
• Anti-Ghosting Safeguard: Automatic 100% payout triggers if 48 hours pass without buyer dispute.

4. How StudentPay Works (5-Step Core Flow)

1. CREATE DEAL: Buyer locks BOT tokens in smart contract escrow.
2. SELLER SUBMITS: Seller submits deliverable & records keccak256 hash on-chain.
3. BUYER INSPECTS: Buyer views watermarked protected preview; original remains locked.
4. ACCEPT OR DISPUTE: Buyer accepts and releases funds, or opens a dispute within 48 hours.
5. DELIVERABLE UNLOCKS: Original downloadable deliverable unlocks post-release.

5. Core Implemented Features

• Blinded Proof Commitment (keccak256 on-chain verification)
• Dual-Level Access Control (Watermarked Preview vs. Original Asset Unlock)
• 48-Hour Review & Dispute Window
• Anti-Ghosting Auto-Release Payouts
• Immediate Deadline Refund Guarantees

6. Technical Architecture & Technology Stack

Frontend: React 18, Vite 6, TypeScript 5.6, wagmi 2.x, viem 2.x, Reown AppKit, TanStack Query.
Blockchain: BotChain Mainnet (Chain ID 677, BOT Native Token), StudentPayEscrow.sol (Solidity 0.8.24).
Testing & Security: 21 passing Foundry/Forge unit & integration tests. Non-custodial, reentrancy-guarded smart contract architecture.

7. Product Status & Roadmap

Current Status: Deployed & Verified V2 Smart Contract (0x59540a9B86c2bFB4F2591c840A6377484200d680) on BotChain Mainnet, complete dApp, 21 passing tests.
Future Roadmap: IPFS decentralized storage integration, multi-token ERC-20 support, partial milestone releases, and cross-chain expansions.